

JERRY JU

☎ +1 (780) 232-5775 | ✉ jerry.ju@uwaterloo.ca | 🏠 jerryjtj.github.io | 🔗 linkedin.com/in/jujerry

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, C, C++, Java, Bash, HTML, CSS
Tools/Frameworks: Git, MySQL, PostgreSQL, MongoDB, Node.js, Express.js, React, Spring, OpenCV, Docker, pandas, Linux, Jenkins, Jira, AWS

EDUCATION

University of Waterloo 2018 – 2023
BASc. in Honours Mechatronics Engineering (*with Computing Option*)
Graduated with Distinction.
Relevant Courses: Data Structures & Algorithms, Computer Structures & Real-Time Systems, Computer Networks, Software Design & Architecture

EXPERIENCE

Q4 Inc. May – Aug. 2022
Full-Stack Developer Intern Toronto, ON

- Built custom React components and hooks in TypeScript for a Fintech SaaS platform serving millions of users, with Jest for unit testing
- Designed REST APIs for event data management using Node.js and MongoDB, enabling scalable storage and real-time updates
- Integrated Twilio APIs to implement real-time chat, increasing customer communication and engagement
- Automated Cypress end-to-end testing and reporting with TestRail and GitHub Actions, streamlining the QA process and reducing production bugs

University of Hawaii Jan. – Apr. 2022
Undergraduate Research Assistant/Software Developer Honolulu, HI

- Developed C++ software to analyze liquid droplets, using OpenGL and computer vision techniques to interpret droplet characteristics
- Collaborated with researchers to integrate embedded software for lab cameras, enabling real-time analysis
- Authored comprehensive technical documentation designed for scientists, researchers and developers

Ford Motor Company May – Aug. 2021
Software Developer Intern Ottawa, ON (remote)

- Optimized Ford's token manager system using C/C++ in Linux environment, improving token security logging and debugging and access for networked vehicles
- Developed cloud-connected, in-vehicle modules on CAN/Ethernet networks, deploying with Jenkins
- Wrote unit and integration tasks in GTest to ensure proper functioning of features

PROJECTS

CivBoards 

- Built a full-stack web app to allow users to parse Civilization VI save games, and to track and record games with other users using a shared platform
- Designed and implemented a Node.js/Express backend with authenticated RESTful CRUD APIs using Auth0, supporting secure data storage and retrieval in a PostgreSQL database

Gymstagram 

- Built a fitness social media app in Java (Android Studio) for workout sharing and user interaction
- Designed a Spring Boot backend with REST APIs and MongoDB for scalable data storage and retrieval

PUBLICATIONS

- J.T.J. Ju, L.N. Boisvert, and Y.Y. Zuo, "Face masks against COVID-19: Standards, efficacy, testing and decontamination methods," *Advances in Colloid and Interface Science*, vol. 292, p. 102435, Jun. 2021. 